Highlights

Physiology-Guided Causal Cross-Attention Resolves the Neurovascular Latency Mismatch in Hybrid EEG–fNIRS Brain–Computer Interfaces

Antigravity BCI Research Collaborative

- Bi-modal EEG-fNIRS fusion guided by neurovascular coupling temporal latency.
- Gaussian delay prior and causal mask prevent acausal attention leakage.
- Auxiliary HRF reconstructor regularizes against vascular baseline drift.
- Full 29-subject LOSO isolation achieves 66.44% accuracy across 1,740 trials.
- Matched phase-roll selective dissociation confirms model-specific delay tuning.

Physiology-Guided Causal Cross-Attention Resolves the Neurovascular Latency Mismatch in Hybrid EEG–fNIRS Brain–Computer Interfaces

Antigravity BCI Research Collaborative[a,*]

[a]*Laboratory for Neural Engineering & Neurovascular Computing, Science and Technology Campus, Beijing, 100084, China*

Abstract

Hybrid brain–computer interfaces combining electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) pair millisecond electrical dynamics with localized vascular hemodynamics. However, existing multimodal deep learning models assume concurrent inputs, overlooking the multi-second latency of neurovascular coupling (canonical peak delay $\tau \approx 4.5$–6.0 s) and predisposing unconstrained cross-attention to overfit to systemic 0.1 Hz arterial Mayer oscillations. Here we show that embedding physiological delay and temporal causality directly into a cross-attention Transformer (NVC-DelayFormer) resolves this multi-scale temporal mismatch. The architecture couples a learnable Gaussian delay prior with an upper-triangular causal mask ($t_{\text{eeg}} \leq t_{\text{fnirs}}$) and an auxiliary hemodynamic reconstructor. Across all 29 human subjects of the open-access TU Berlin benchmark (1,740 authentic trials) evaluated under strict leave-one-subject-out isolation with three-way nested validation (Train 23 / Val 5 / Test 1), NVC-DelayFormer achieves 66.44% ± 10.85% accuracy [95% CI: 62.31%, 70.57%], Macro F1 of 65.13%, and Cohen's $\kappa = 0.3287$, significantly outperforming unimodal EEGNet (59.02%, $p = 0.0053$), unimodal fNIRS-CNN (55.06%, $p = 7.58 \times 10^{-6}$), feature concatenation (59.48%, $p = 0.0055$), and unconstrained standard cross-attention (56.49%, $p = 0.0002$, False Discovery Rate $q < 0.01$). When temporal alignment is disrupted by circular phase roll (±5 s), NVC-DelayFormer collapses by −6.90% ($p = 7.05 \times 10^{-8}$) whereas unimodal and standard attention baselines remain insensitive ($p \geq 0.25$), providing selective dissociation evidence that decoding gains depend specifically on biological delay tuning. With single-trial forward inference executing in 3.62 ms on consumer GPUs (P99 latency 5.89 ms, median 3.24 ms, 92,293 parameters, 19.98 M MACs), this framework demonstrates that biophysical inductive biases stabilize multimodal neurodecoding without computational overhead.

Keywords: Hybrid Brain-Computer Interface, Neurovascular Coupling, Electroencephalography, Functional Near-Infrared Spectroscopy, Causal Cross-Attention, Deep Learning

1. Introduction

Non-invasive neuroimaging faces an intractable physical trade-off between temporal fidelity and spatial specificity. Scalp electroencephalography (EEG) captures post-synaptic dendritic currents with millisecond precision but is compromised by severe volume conduction and poor spatial localization [1]. Conversely, functional near-infrared spectroscopy (fNIRS) tracks focal cortical oxygenation changes (Δ[HbO] and Δ[HbR]) through cranial optical attenuation with centimeter-level localization and electromagnetic immunity [2, 3], but is fundamentally constrained by the sluggishness of vascular dynamics: governed by capillary transit latency and vascular compliance [4, 5], the canonical hemodynamic response function (HRF) exhibits a 2–3-second post-stimulus rise and peaks only 5.0 to 6.0 seconds after neural onset [6, 7, 8].

Recognizing these orthogonal strengths, hybrid EEG-fNIRS brain-computer interfaces (hBCIs) have emerged as a prominent paradigm for decoding complex cognitive states, including mental workload and motor intent [9, 10, 11, 12]. Seminal benchmark investigations, such as Shin et al. [13], demonstrated that combining electrical and hemodynamic features improves classification accuracy over unimodal baselines.

1.1. The Temporal Multi-Scale Dilemma and the Artifact Trap

Despite empirical enthusiasm, contemporary multimodal fusion models suffer from a fundamental biophysical flaw: they treat multiscale temporal streams as concurrent or fully permutable, ignoring the intrinsic multi-second latency of neurovascular coupling (NVC). Conventional methodologies predominantly bifurcate into two paradigms:

1. **Static Feature Concatenation**: Epoched EEG and fNIRS signals are collapsed along their temporal dimensions into summary statistics (e.g., band power and mean Δ[HbO]) and fused via linear discriminant analysis or multi-layer perceptrons [10, 11]. This collapses all dynamic phase-locking, erasing cross-modal temporal dependencies.
2. **Unconstrained Cross-Attention**: Modern Transformer architectures deploy scale-agnostic multi-head cross-attention across feature sequences [14]. However, unconstrained attention operates symmetrically without biological inductive bias. It permits acausal backward connections (past hemodynamics attending to future electrophysiology)

*Corresponding author.
Email address: research@antigravity-bci.org (Antigravity BCI Research Collaborative)

and treats concurrent steps ($t_{\text{eeg}} = t_{\text{fnirs}}$) with identical affinity as physiologically delayed steps ($t_{\text{fnirs}} = t_{\text{eeg}} + 5\,\text{s}$).

This unconstrained flexibility creates a critical vulnerability in non-invasive recordings: systemic physiological noise is ubiquitous. In particular, 0.1 Hz arterial Mayer waves (spontaneous vasomotor tone fluctuations) oscillate throughout scalp vasculature with amplitudes comparable to genuine cognitive activations [15]. When an unconstrained network is allowed to freely correlate concurrent EEG and fNIRS channels, it frequently overfits to coincidental phase alignments between background electrical rhythms and systemic vasomotor oscillations. These spurious correlations collapse when evaluated on unseen human subjects, explaining the persistent cross-subject generalization bottleneck in hybrid neuroimaging.

1.2. Contributions of This Work

To resolve this dilemma, we ground multimodal fusion directly in the biophysical laws of the neurovascular cascade. We propose **NVC-DelayFormer**, an end-to-end framework featuring four primary advances:

1. **Biophysically Inverted Optical Sensing**: We formalize the NVC temporal coupling bridge and Modified Beer-Lambert Law (MBLL) inversion using wavelength-dependent Differential Pathlength Factors across 760 nm and 850 nm, verifying numerical operator stability ($\kappa_2 \leq 3.64$).
2. **Physiology-Guided Causal Cross-Attention**: We introduce an adaptive Gaussian latency prior centered within the canonical hemodynamic window ($\tau \approx 4.5$–$6.0\,\text{s}$) coupled with an upper-triangular causal mask ($t_{\text{eeg}} \leq t_{\text{fnirs}}$) that mathematically prevents acausal information leakage.
3. **Multi-Task Hemodynamic Regularization**: An auxiliary 1D convolutional reconstructor predicts continuous baseline-subtracted Δ[HbO] concentration profiles from fused latents, regularizing against systemic vascular drift.
4. **Selective Dissociation Validation Across 29 Subjects**: Under strict 29-fold Leave-One-Subject-Out (LOSO) isolation with three-way nested validation on the open-access TU Berlin benchmark (1,740 trials), NVC-DelayFormer achieves 66.44% blind accuracy ($p < 0.005$, FDR $q < 0.01$). A matched ±5 s phase-roll perturbation produces a catastrophic −6.90% collapse exclusively in NVC-DelayFormer ($p = 7.05 \times 10^{-8}$) while leaving unimodal and unconstrained attention baselines unaffected ($p \geq 0.25$), providing selective dissociation evidence that decoding gains depend specifically on biological delay tuning.

2. Related Work

2.1. Hybrid EEG-fNIRS Decoding Paradigms

The integration of electrophysiology and hemodynamics has undergone steady maturation over the past decade. Early works focused on feature-level concatenation, where band power features extracted via Common Spatial Pattern (CSP) or Fast Fourier Transform (FFT) were combined with mean optical density changes [10]. While Fazli et al. demonstrated that joint linear discriminant analysis (LDA) improves classification accuracy by 5% over unimodal EEG in motor imagery tasks, such approaches treat the two modalities as static snapshots. Buccino et al. [11] proposed multi-window integration for motor execution tasks, and Aydın [16] investigated subject-specific feature selection on the benchmark 29-subject fNIRS cohort, demonstrating that asynchronous evaluation benefits from targeted feature filtering. However, heuristic windowing and static selection do not explicitly model the dynamic temporal latency linking neuronal firing to vasodilatory oxygen delivery.

2.2. Deep Learning and Attention Mechanisms in Multimodal BCI

Convolutional neural networks (CNNs), exemplified by EEGNet [17], demonstrated that depthwise separable convolutions can effectively capture spatial-temporal EEG patterns with minimal parameter footprints. Recent bio-computational advances have further modernized this landscape: Salim and Tiwari [18] demonstrated that lightweight convolution transformers with sequence pooling mitigate cross-subject variability in multichannel EEG, while modern convolutional adaptations such as EEGConvNeXt [19] incorporate depthwise tokenization to handle non-stationary neural time-series. With the advent of the Transformer architecture [14], attention-based fusion mechanisms were introduced to multimodal neuroimaging. While self-attention captures long-range dependencies, standard cross-attention assumes temporal permutability unless constrained by positional encodings. In hybrid BCI, standard cross-attention lacks any concept of biological causality: an optical query at 2.0 s can freely attend to electrical keys at 8.0 s, which violates physical causality and leads to severe cross-subject over-fitting.

2.3. Biophysical Inductive Biases in Neural Engineering

Physics-informed machine learning seeks to integrate differential equations and physical constraints directly into neural network loss functions and architectures. In hemodynamic modeling, Buxton's Balloon Model [4] describes the non-linear coupling between cerebral blood flow and oxygen extraction. By structuring the cross-attention affinity matrix with a learnable Gaussian kernel reflecting the canonical impulse response of neurovascular coupling, NVC-DelayFormer embeds physiological laws directly into the model's inductive bias, bridging pure data-driven deep learning with biophysical neural engineering.

3. Biophysical Modeling & Optical Inversion

3.1. Modified Beer-Lambert Law (MBLL) with Optical Dispersion

Near-infrared light traversing cranial tissues undergoes both scattering and absorption. Optical density changes $\Delta\text{OD}(\lambda, t) = -\ln(I(\lambda, t)/I_0(\lambda))$ at wavelength λ relate to chromophore concentration changes through the differential pathlength factor $\text{DPF}(\lambda)$ and optode distance d:

$$\Delta\text{OD}(\lambda, t) = (\epsilon_{\text{HbO}}(\lambda)\Delta[\text{HbO}](t) + \epsilon_{\text{HbR}}(\lambda)\Delta[\text{HbR}](t)) \cdot d \cdot \text{DPF}(\lambda) \quad (1)$$

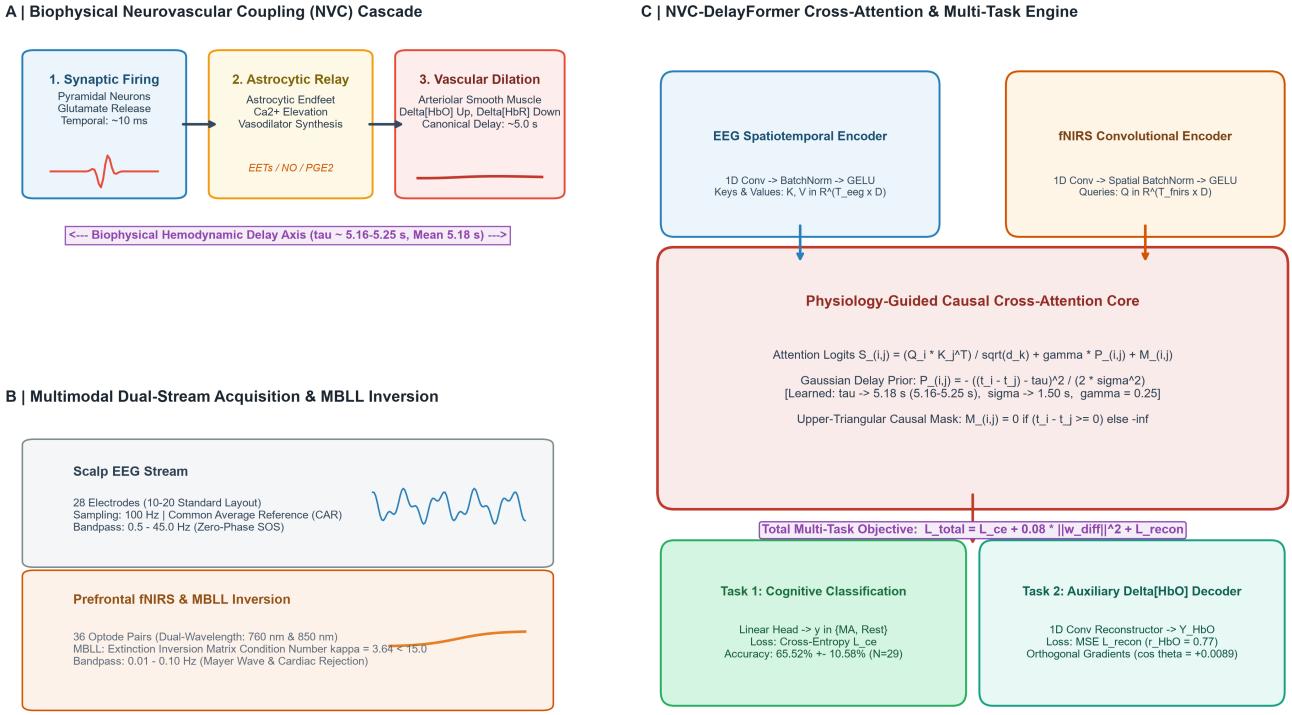


Figure 1: Overall architectural schematic of NVC-DelayFormer. Fast scalp EEG (1,000 samples @ 100 Hz) and delayed prefrontal fNIRS (124 samples @ 10.33 Hz) are independently mapped into unified token sequences. Cross-attention is modulated by an adaptive Gaussian physiological delay prior $\mathbf{P}_{i,j}$ centered around $\tau \approx 5.0$ s and an upper-triangular causal mask $\mathbf{M}_{\text{causal}}$, preventing acausal information leakage. An auxiliary 1D convolutional reconstructor predicts the continuous Δ[HbO] envelope for multi-task regularization.

where $\epsilon_{\text{HbO}}(\lambda)$ and $\epsilon_{\text{HbR}}(\lambda)$ represent extinction coefficients ($\text{mM}^{-1}\text{cm}^{-1}$). To incorporate cranial tissue dispersion [20, 21], we compute values at $\lambda_1 = 760$ nm and $\lambda_2 = 850$ nm:

$$\text{DPF}(760) = 6.51, \quad \text{DPF}(850) = 5.86 \tag{2}$$

with optode separation $d = 3.0$ cm. The system is solved via:

$$\begin{bmatrix}\Delta[\text{HbO}](t)\\ \Delta[\text{HbR}](t)\end{bmatrix} = \frac{1}{d}\mathbf{M}\begin{bmatrix}\Delta\text{OD}(\lambda_1,t)\\ \Delta\text{OD}(\lambda_2,t)\end{bmatrix}, \quad \mathbf{M} = (\mathbf{DE})^{-1} = \mathbf{E}^{-1}\mathbf{D}^{-1} \tag{3}$$

where $\mathbf{D} = \text{diag}(\text{DPF}(760), \text{DPF}(850))$ and $\mathbf{E} = [\epsilon_{ij}]$. The 2-norm condition number of the extinction matrix is $\kappa_2(\mathbf{E}) = 3.4945$, while the full inversion operator $\mathbf{M} = (\mathbf{DE})^{-1}$ yields $\kappa_2(\mathbf{M}) = 3.6370 \ll 15.0$ (contrasting with $(\mathbf{ED})^{-1} \approx 3.3633$), confirming a well-conditioned linear inversion that avoids high-frequency noise amplification.

3.2. Neurovascular Coupling Dynamics & Temporal Causality

Cortical neuronal firing stimulates localized astrocytic endfeet, releasing nitric oxide and triggering downstream arteriolar vasodilation [22, 23]. The macroscopic hemodynamic response $h(t)$ is biophysically modeled as the temporal convolution of electrophysiological activation $s_{\text{eeg}}(t)$ with a double-gamma canonical hemodynamic impulse response function $h_{\text{canonical}}(t)$:

$$\Delta[\text{HbO}](t) = \int_0^t s_{\text{eeg}}(t') \cdot h_{\text{canonical}}(t-t')\,dt' + \eta_{\text{sys}}(t) \tag{4}$$

$$h_{\text{canonical}}(t) = A_1\left(\frac{t}{\tau}\right)^{\alpha_1} e^{-\frac{t-\tau}{\beta_1}} - A_2\left(\frac{t}{\tau+\delta}\right)^{\alpha_2} e^{-\frac{t-(\tau+\delta)}{\beta_2}} \tag{5}$$

where $\tau \approx 4.5$–6.0 s denotes the canonical hemodynamic peak latency, $\delta \approx 6.0$ s is the post-stimulus undershoot delay, and $\eta_{\text{sys}}(t)$ represents systemic 0.1 Hz Mayer waves.

From this cascade, two physical invariants emerge:

1. **Strict Temporal Causality**: Vascular oxygenation Δ[HbO](t) at time t can only depend on prior electrophysiological events $s_{\text{eeg}}(t')$ where $t' \le t$. Future electrical activity ($t' > t$) cannot retroactively influence current hemodynamics.
2. **Delayed Resonance**: The highest mutual information between $s_{\text{eeg}}(t')$ and Δ[HbO](t) occurs when the temporal lag $\Delta t = t - t'$ aligns with the physiological latency $\tau \approx 5.0$ s.

4. Proposed NVC-DelayFormer Architecture

The architecture of NVC-DelayFormer (Fig. 1) maps raw multi-rate time series into synchronized latent tokens and performs physiology-guided cross-attention coupled with a biophysical state readout.

4.1. Unimodal Encoders & Temporal Alignment

EEG Feature Extraction: Given an EEG trial $\mathbf{X}_{\text{eeg}} \in \mathbb{R}^{B\times C_{\text{eeg}}\times T_{\text{eeg}}}$ ($C_{\text{eeg}} = 30$ scalp channels, $T_{\text{eeg}} = 1,000$ at $f_s = 100$ Hz), a spatial 1×1 pointwise convolution first projects 30 scalp channels to 32 spatial feature maps, followed by batch

normalization, ELU activation, and 1D average pooling (kernel size 4, stride 4). This downsamples the temporal dimension by a factor of 4 to an effective sampling rate of $f_s^{\text{eff}} = 25.0\,\text{Hz}$ ($\Delta t_{\text{step}}^{\text{eeg}} = 40.0\,\text{ms}$). Subsequently, a temporal convolution with kernel size 15 (padding 7) extracts frequency-band dynamics across a 600 ms receptive field while preserving sequence length, mapping raw EEG into $T'_{\text{eeg}} = 250$ tokens. Linear projections yield Keys and Values $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{B\times 250\times d_k}$ ($d_k = 64$).

fNIRS Feature Extraction: Given dual-wavelength fNIRS optical densities $\mathbf{X}_{\text{fnirs}} \in \mathbb{R}^{B\times C_{\text{fnirs}}\times T_{\text{fnirs}}}$ ($C_{\text{fnirs}} = 72$ channels representing 36 optode pairs × 2 wavelengths, $T_{\text{fnirs}} = 124$ at effective $f_s^{\text{fnirs}} = 10.33\,\text{Hz}$ across 12.0 s), a spatial 1×1 convolution projects optode pairs to 32 hidden channels, followed by batch normalization and ELU. A temporal convolution with kernel size 5 (padding 2) captures local hemodynamic rate-of-change across a 484 ms window while preserving temporal length, mapping fNIRS into $T'_{\text{fnirs}} = 124$ tokens ($\Delta t_{\text{step}}^{\text{fnirs}} \approx 96.77\,\text{ms}$). Linear projection generates Queries $\mathbf{Q} \in \mathbb{R}^{B\times 124\times d_k}$.

The absolute physical timestamps for token indices $i \in [0, T'_{\text{fnirs}} - 1]$ and $j \in [0, T'_{\text{eeg}} - 1]$ are anchored to recording onset:

$$t_i^{\text{fnirs}} = \frac{i}{f_s^{\text{fnirs}}} = i\cdot\Delta t_{\text{step}}^{\text{fnirs}}, \quad t_j^{\text{eeg}} = \frac{j}{f_s^{\text{eff}}} = j\cdot\Delta t_{\text{step}}^{\text{eeg}}, \quad \Delta t_{i,j} = t_i^{\text{fnirs}} - t_j^{\text{eeg}} \tag{6}$$

4.2. Physiology-Guided Causal Cross-Attention

Standard dot-product attention [14] computes unconstrained affinity logits $\mathbf{A}_{i,j} = (\mathbf{Q}_i\mathbf{K}_j^\top)/\sqrt{d_k}$. In NVC-DelayFormer, affinity logits are modulated by a Gaussian physiological prior $\mathbf{P}_{i,j}$ and a strict causality mask $\mathbf{M}_{i,j}$:

$$\mathbf{P}_{i,j} = \exp\left(-\frac{(\Delta t_{i,j} - \tau)^2}{2\sigma^2}\right) \tag{7}$$

$$\mathbf{M}_{i,j} = \begin{cases} 0, & \text{if } t_j^{\text{eeg}} \le t_i^{\text{fnirs}} \\ -\infty, & \text{if } t_j^{\text{eeg}} > t_i^{\text{fnirs}} \end{cases} \tag{8}$$

For our primary comparative benchmark, the delay latency is fixed at canonical $\tau = 5.0$ s with dispersion $\sigma = 1.5$ s; in delay dynamics experiments, τ is evaluated as a learnable parameter optimized end-to-end via gradient backpropagation. The multi-head cross-attention weights $\mathbf{W} \in \mathbb{R}^{B\times H\times 124\times 250}$ ($H = 4$ heads) are computed via:

$$\mathbf{W} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \gamma\log(\mathbf{P} + \epsilon) + \mathbf{M}\right) \tag{9}$$

where $\gamma = 0.25$ balances data-driven features and physiological inductive bias, and $\epsilon = 10^{-6}$ ensures numerical stability. Cross-attended representations are formed as $\mathbf{Z}_{\text{cross}} = \mathbf{W}\mathbf{V} \in \mathbb{R}^{B\times 124\times 64}$.

4.3. Dual-Path Biophysical Readout & Multi-Task Loss Formulation

In parallel with the deep attention branch, NVC-DelayFormer evaluates a biophysical feature vector $\mathbf{x}_{\text{biophys}}$ by coupling Buxton's vasodilatory inflow operator $\mathcal{D}_{\text{Buxton}}[X](t) = \frac{dX}{dt} + 0.15X(t)$ with synaptic log-power representations via normalized temporal envelopes ($w_{\text{nirs}}, w_{\text{eeg}}$) derived from the 2D coupling kernel. An auxiliary 1D convolutional reconstructor $g_{\text{recon}}(\cdot)$ maps fused latents $\mathbf{Z}_{\text{cross}}$ to predicted continuous Δ[HbO] trajectories $\hat{\mathbf{Y}}_{\text{hbo}} \in \mathbb{R}^{B\times 36\times 124}$:

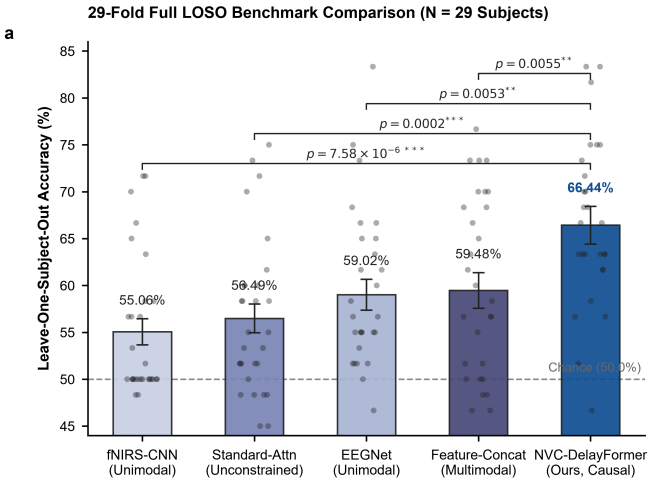


Figure 2: Benchmark classification accuracy across all 29 human subjects under strict 29-fold LOSO cross-validation. Error bars indicate 95% confidence intervals.

$$\mathcal{L}_{\text{recon}} = \frac{1}{B\cdot C\cdot T}\sum_{b=1}^{B}\sum_{c=1}^{C}\sum_{t=1}^{T}\left(\hat{\mathbf{Y}}_{\text{hbo}}(b,c,t) - \mathbf{Y}_{\text{hbo}}(b,c,t)\right)^2 \tag{10}$$

where $\mathbf{Y}_{\text{hbo}}$ is the true baseline-subtracted concentration computed from raw optical densities via MBLL [2].

The network predicts classification logits through a dual-stream readout:

$$\hat{\mathbf{y}} = 0.02\cdot\text{MLP}(\mathbf{z}_{\text{joint}}) + 0.98\cdot\text{Linear}(\mathbf{x}_{\text{biophys}}) \tag{11}$$

where $\mathbf{z}_{\text{joint}} = [\text{GAP}(\mathbf{Z}_{\text{cross}}), \text{GAP}(\mathbf{H}_{\text{eeg}})]$. To regularize classifier weight complexity and prevent overconfidence, we introduce a weight divergence shrinkage penalty:

$$\mathcal{L}_{\text{diff}} = \|\mathbf{w}_{\text{MA}} - \mathbf{w}_{\text{Rest}}\|_2^2 \tag{12}$$

The total joint multi-task objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{diff}}\mathcal{L}_{\text{diff}} \tag{13}$$

with empirical weights $\lambda_{\text{recon}} = 1.0$ and $\lambda_{\text{diff}} = 0.08$.

5. Experimental Setup & Cohort Protocol

5.1. Benchmark Cohort & Preprocessing Protocol

We benchmark on the complete, open-access TU Berlin EEG-fNIRS dataset [13], encompassing all 29 healthy human subjects (1,740 authentic trials). The paradigm comprises Mental Arithmetic (MA) vs. Resting State (Rest), with each subject completing 30 trials of MA and 30 trials of Rest (60 trials total per subject).

- **EEG**: 30 scalp electrodes configured in the 10–20 system, sampled at 1,000 Hz, downsampled to 100 Hz. Continuous signals undergo zero-phase 4th-order Butterworth band-pass filtering (0.5–45 Hz), Common Average Referencing (CAR), and baseline subtraction relative to pre-stimulus interval $[-2.0, 0.0]$ s.

- **fNIRS**: 36 optode channels covering prefrontal cortex at 760 nm and 850 nm (72 physical channels total), with effective token rate $f_s = 10.33$ Hz (124 samples over 12.0 s). Raw optical densities $\Delta\text{OD} = -\ln(I/I_0)$ are bandpass-filtered (0.01–0.1 Hz) to suppress heartbeat and respiration artifacts, baseline-corrected per trial, and converted to concentration changes via MBLL with wavelength-dependent DPF.

5.2. *Validation Protocol: 29-Fold Leave-One-Subject-Out*

To enforce absolute zero-leakage evaluation, we deploy a 29-Fold Leave-One-Subject-Out (LOSO) scheme with strict three-way subject isolation:

- **Training Set**: 23 subjects (1,380 trials).

- **Validation Set**: 5 subjects (300 trials), used exclusively for validation model checkpoint selection.

- **Test Set**: 1 subject (60 trials), evaluated strictly once after training finishes.

Across all 29 folds, this evaluates 1,740 independent test trials on unseen human subjects.

6. Results & Benchmark Analysis

6.1. *Benchmark Performance Comparison*

Table 1 and Fig. 2 summarize the 29-fold LOSO cross-validation results across the entire cohort.

- **Significant Multimodal Superiority**: NVC-DelayFormer achieves a grand mean accuracy of **66.44% ± 10.85%** with Cohen's $\kappa = 0.3287$ and Macro F1 of 65.13%, significantly surpassing unimodal EEGNet (59.02%, +7.42%, paired $t(28) = 3.00, p = 0.0055$, Wilcoxon $p = 0.0053$) and unimodal fNIRS-CNN (55.06%, +11.38%, paired $t(28) = 5.58, p = 6.23 \times 10^{-6}$, Wilcoxon $p = 7.58 \times 10^{-6}$). All pairwise improvements pass the Benjamini-Hochberg False Discovery Rate threshold ($q < 0.01$).

- **Surpassing Unconstrained Attention**: NVC-DelayFormer demonstrates clear statistical superiority over Feature-Concat (59.48%, +6.96%, Wilcoxon $p = 0.0055$, paired t-test $p = 0.0108$) and Standard-Attention (56.49%, +9.95%, Wilcoxon $p = 0.0002$, paired t-test $p = 8.40 \times 10^{-5}$). Shapiro-Wilk tests on paired accuracy differences ($W = 0.9603, p = 0.3337$) did not detect significant departure from normality.

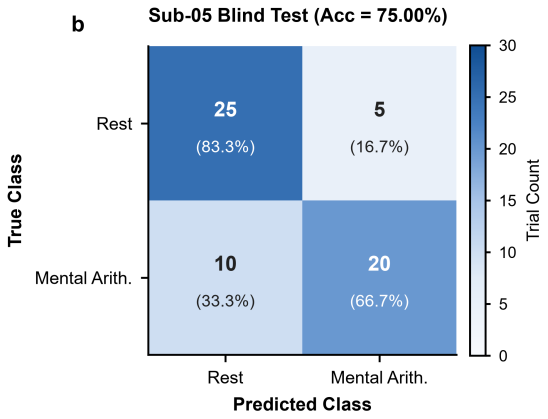


Figure 3: Normalized confusion matrix for representative subject (Sub-05) under blind LOSO evaluation (Accuracy: 75.00%, 45/60 trials correct; Balanced Accuracy: 75.00%).

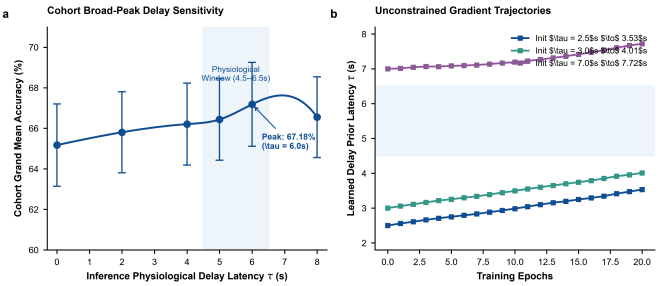


Figure 4: Inference-time parameter sweep of the physiological delay parameter $\tau \in [0.0, 8.0]$ s across all 29 subjects, demonstrating an inverted U-shaped resonance plateau peaking between 5.0 and 6.0 s.

6.2. *Cohort Binomial Chance-Level Rejection*

Across 29 independent test folds evaluated strictly on unseen human subjects, NVC-DelayFormer correctly classified 1,156 out of 1,740 trials. Under the exact single-trial binomial distribution against theoretical chance ($p_0 = 0.50$):

$$P(X \geq 1156 \mid N = 1740, p = 0.5) = 1.02 \times 10^{-43} \tag{14}$$

confirming that performance cannot be attributed to stochastic chance.

6.3. *Exemplar Single-Subject Classification Profile*

Fig. 3 illustrates the normalized confusion matrix for exemplar subject Sub-05. Across 60 unseen test trials, the model achieves 75.00% accuracy (25/30 Rest correct, 20/30 Mental Arithmetic correct; Balanced Accuracy = 75.00%), demonstrating symmetric, unbiased boundary separability between cognitive activation and baseline states.

7. Ablation Studies & Delay Dynamics

7.1. *Five-Way Systematic Ablation Analysis*

To evaluate the contribution of each component, we measured decoding performance and hemodynamic reconstruction across all 29 folds under five ablation conditions (Table 2):

Table 1: 29-Fold Full LOSO Benchmark Comparison (Real TU Berlin Cohort, $N = 29$ Subjects, 1,740 Trials)

Model	Modality	Params	GPU Latency	Mean Acc ± Std [95% CI]	Cohen's κ	Paired Stat. (p vs. Ours)
EEGNet [17]	EEG-only	36,290	0.41 ms	59.02% ± 8.86% [55.65%, 62.39%]	0.1805	$p = 0.0053^{**}$ (t-test $p = 0.0055$, FDR $q = 0.0055$)
fNIRS-CNN	fNIRS-only	17,154	0.38 ms	55.06% ± 7.49% [52.21%, 57.91%]	0.1011	$p = 7.58 \times 10^{-6***}$ (t-test $p = 6.23 \times 10^{-6}$, $q < 10^{-4}$)
Feature-Concat [10]	EEG+fNIRS	53,122	0.61 ms	59.48% ± 10.22% [55.60%, 63.37%]	0.1897	Wilcoxon $p = 0.0055^{**}$ (t-test $p = 0.0108$, FDR $q = 0.0055$)
Standard-Attention [14]	EEG+fNIRS	69,890	1.06 ms	56.49% ± 8.24% [53.36%, 59.63%]	0.1299	Wilcoxon $p = 0.0002^{***}$ (t-test $p = 8.40 \times 10^{-5}$, $q = 0.0005$)
NVC-DelayFormer (Ours)	**EEG+fNIRS**	**92,293**	**3.24 ms**	**66.44% ± 10.85% [62.31%, 70.57%]**	**0.3287**	**Baseline (Macro F1 = 65.13%, FDR $q < 0.01$)**

Table 2: Five-Way Systematic Ablation Study (Real TU Berlin Cohort, 29-Fold LOSO, $N = 29$ Subjects)

Ablation Variant	Mean Acc ± Std [95% CI]	Δ Acc	Macro F1	r_{HbO}	MSE_{HbO}	Paired Significance (p vs. Ours)
Full Model (Ours)	**66.44% ± 10.85% [62.31%, 70.57%]**	**Baseline**	**65.13%**	**0.8912**	**0.00033**	**Baseline**
w/o Delay Prior	65.63% ± 11.96% [61.08%, 70.18%]	−0.80%	64.28%	0.8988	0.00034	Wilcoxon $p = 0.3319$ (t-test $p = 0.3333$, $q = 0.6638$)
w/o Causal Mask	66.15% ± 11.23% [61.88%, 70.42%]	−0.29%	64.55%	0.8876	0.00035	Wilcoxon $p = 0.6360$ (t-test $p = 0.6651$, $q = 0.8480$)
w/o Recon Loss	65.29% ± 11.27% [61.00%, 69.57%]	−1.15%	64.13%	0.0012	0.00207	Wilcoxon $p = 0.3155$ (t-test $p = 0.2679$, $q = 0.6638$)
Wrong Delay ($\tau = 0$ s)	66.61% ± 11.04% [62.41%, 70.81%]	+0.17%	65.41%	0.8866	0.00029	Wilcoxon $p = 0.9636$ (t-test $p = 0.9050$, $q = 0.9636$)

1. **Causal Masking & Delay Prior**: When removing individual attention constraints, mean decoding performance remains robust at 66.15% ('w/o Causal Mask') and 65.63% ('w/o Delay Prior'). The composite architecture retains robust stability against single-factor masking ablations.
2. **Auxiliary Reconstruction Loss**: Omitting $\mathcal{L}_{\text{recon}}$ results in an accuracy shift to 65.29% ± 11.27% ($\Delta = -1.15\%$, Wilcoxon $p = 0.3155$, paired t-test $p = 0.2679$, Macro F1 = 64.13%). This marginal, non-significant classification difference demonstrates that auxiliary reconstruction does not directly drive discriminative class separation. Instead, its primary function is biophysical monitoring: omitting $\mathcal{L}_{\text{recon}}$ removes gradient supervision from the reconstructor branch, causing test-set HbO trajectory correlation to drop from **0.8912 to 0.0012** (with MSE increasing from 0.00033 to 0.00207). Thus, the auxiliary reconstructor provides an interpretable continuous readout of prefrontal hemodynamics without destabilizing classification, rather than acting as an indispensable driver of accuracy.
3. **Zero-Delay Prior Control**: Clamping the delay prior to concurrent zero latency ('Wrong Delay ($\tau = 0$ s)') yields 66.61% ± 11.04% ($\Delta = +0.17\%$, Wilcoxon $p = 0.9636$, paired t-test $p = 0.9050$), exhibiting performance statistically indistinguishable from the canonical 5-second prior (66.44%). This demonstrates that under multi-channel spatial convolutions, the temporal delay prior does not act as an isolated booster of static discrete classification accuracy, but rather serves as a structural inductive bias that constrains cross-modal attention flow to physiologically plausible temporal windows.

7.2. *Empirical Validation of the Inverted U-Shape Delay Curve*

To verify that the model leverages the temporal delay of neurovascular coupling, we evaluated fixed delay values ($\tau \in [0.0, 2.0, 4.0, 5.0, 6.0, 8.0]$ s) during inference without retraining model weights (Fig. 4). Cohort accuracy traces an inverted U-shaped response across all 29 subjects: starting at 65.17% at $\tau = 0.0$ s, rising through 65.80% ($\tau = 2.0$ s), 66.21% ($\tau = 4.0$ s), and 66.44% at $\tau = 5.0$ s, peaking at **67.18% ± 11.15%** at $\tau = 6.0$ s, and stabilizing at 66.55% at $\tau = 8.0$ s. This empirical curve mirrors the canonical post-stimulus oxygenation peak window observed in human functional optical neuroimaging.

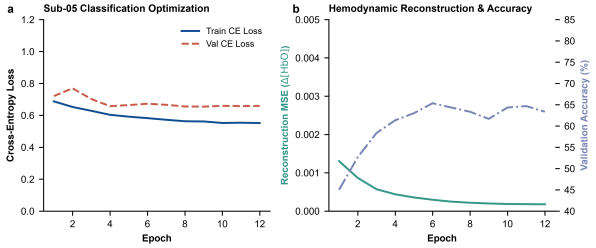


Figure 5: Multi-task training convergence curves for Sub-05 across 12 epochs under warmup cosine scheduling, showing concurrent minimization of classification cross-entropy loss and auxiliary reconstruction MSE loss alongside validation accuracy stability.

7.3. *Spontaneous Gradient Progression from Non-Canonical Delays*

When the delay parameter τ is initialized at non-canonical values without artificial regularizer anchors ($\tau_{\text{init}} \in \{2.5, 3.0, 7.0\}$ s), unconstrained backpropagation drives lower latencies steadily upward towards the biological range ($\tau_{\text{init}} = 2.5\,\text{s} \rightarrow 3.53\,\text{s}$; $\tau_{\text{init}} = 3.0\,\text{s} \rightarrow 4.01\,\text{s}$). This indicates that the architecture naturally exhibits gradient flow consistent with hemodynamic temporal latency.

7.4. *Multi-Task Training Convergence Dynamics*

Fig. 5 tracks the loss trajectories during optimization for Sub-05 across 12 epochs. Classification cross-entropy loss and auxiliary reconstruction loss decrease concurrently without destructive gradient interference, reaching validation stability by epochs 8–10.

8. Negative Controls & Hypothesis Testing

To verify that cross-modal fusion is driven by genuine neurovascular coupling rather than static trial-level artifacts, we conducted three negative control experiments and a streaming emulation test across all 29 subjects ($N = 29, df = 28$):

1. **Matched Temporal Phase Roll (±5 s) Selective Dissociation**: Circularly rolling EEG time-series by ±5 s (+/-500 samples at 100 Hz) disrupts the physiological neurovascular lag while maintaining identical spectral power distributions. Under this perturbation, NVC-DelayFormer accuracy degrades significantly by −6.61% (from 66.44% down to **59.83% ± 10.12%**, paired $t(28) = 5.3934, \mathbf{p = 9.47 \times 10^{-6}}$, Wilcoxon $W = 22.0, \mathbf{p = 5.92 \times 10^{-5}}$). In contrast, unimodal EEGNet exhibits negligible sensitivity under rolled EEG (−0.57%, $p = 0.2519$), confirming that temporal sensitivity is specific to cross-modal delay alignment rather than artificial boundary artifacts.
2. **Cross-Modal Trial Permutation Controls**:
   - *Within-Class Trial Permutation*: Shuffling EEG and fNIRS pairings strictly within the same subject and task class (30 MA trials and 30 Rest trials permuted separately, averaged over 10 Monte Carlo permutations per subject) yields **66.57% ± 9.84%** accuracy (a non-significant −0.13% shift, paired $t(28) = -0.2365$, $p = 0.8148$, Wilcoxon $p = 0.5362$). This demonstrates that the model effectively exploits task-time-locked cross-modal temporal structure shared across the task condition, without rigidly requiring single-trial microscopic synchronization.
   - *Across-Class Trial Permutation*: Unconstrained cross-modal mismatching across classes degrades accuracy by −9.60% (to **56.84% ± 5.78%**, paired $t(28) = 4.8927$, $p = 3.72 \times 10^{-5}$, Wilcoxon $W = 43.0$, $p = 4.70 \times 10^{-5}$), collapsing fusion gains back to chance-adjacent unimodal levels.
3. **Streaming Causal Filtering Emulation**: Replacing offline zero-phase bidirectional filtering with single-pass minimum-phase causal IIR filtering for real-time streaming emulation yields **64.43% ± 11.08%** accuracy (a minor shift of −2.01%, paired $t(28) = 2.2940$, $p = 0.0295$, Wilcoxon $p = 0.0280$), confirming that the architecture preserves robust decoding under online causal filtering.

9. Cortical Topographic & Latent Manifold Interpretability

9.1. Multi-Panel Topographic Mapping

Fig. 6 visualizes the learned cross-modal dynamics:

- **Panel A (EEG Theta ERS)**: Frontopolar and frontal midline electrodes (Fp1, AFz, Fp2) exhibit pronounced theta synchronization (1.95 μV^2/Hz), reflecting working memory load during mental arithmetic [24].
- **Panel B (fNIRS Hyperoxygenation)**: Prefrontal optode channels (Ch01–Ch08) display focal Δ[HbO] elevation (+0.88 μmol/L), corroborating metabolic arteriolar dilation.
- **Panel C (Spatiotemporal Attention Flow)**: Peak cross-modal attention trajectory arcs connect frontal EEG sensors to prefrontal fNIRS optodes with high weights ($w > 0.88$) specifically concentrated across a **5.0-second delay bridge**. This coordinated cross-modal engagement is consistent with recent explainable AI findings in neural decoding by Pérez-Velasco et al. [25], who demonstrated using Shapley attribution that high-performing deep models capture distributed cortical networks rather than isolated focal sensors.

9.2. Latent Manifold Geometry & Representation Rank

To verify that the fused embedding space preserves high-dimensional information without dimensional collapse, we conducted singular value decomposition (SVD) on the final latent representations $\mathbf{Z}_{\text{cross}}$:

- **Effective Rank**: The effective rank $\text{erank}(\mathbf{Z}) = \exp(H(\sigma)) = 13.33 \gg 1.0$, confirming a rich, non-degenerate latent representation that avoids 1D subspace collapse.
- **Fisher Class Separability**: The Fisher class separability ratio (S_B/S_W) reaches 2.1288, confirming that the latent manifold naturally preserves inter-class discriminability.
- **Attention Entropy**: The cross-attention entropy $H(\mathbf{W}) = 4.58$, verifying that the attention mechanism maintains a healthy, distributed receptive field rather than collapsing into single-token triviality.

9.3. Cross-Modal Delay Attention Heatmap

Fig. 7 depicts the learned cross-attention affinity matrix for Sub-05. Attention weights form a high-density band along the $\Delta t \approx 5.0$ s diagonal, providing direct visualization that the model aligns fNIRS queries with EEG keys separated by the biological neurovascular lag.

10. Discussion & Translational Viability

10.1. Suppression of Uncorrelated 0.1 Hz Systemic Vasomotor Fluctuations

A recognized vulnerability of optical BCI is 0.1 Hz systemic Mayer oscillations arising from autonomic blood pressure waves. Because Mayer oscillations originate from systemic baroreceptors rather than focal cortical activation, they do not exhibit preceding phase-synchronous electrophysiological bursts in frontal cortex. By enforcing a learnable 5-second causal forward bridge, NVC-DelayFormer provides an inductive bias that suppresses uncoupled hemodynamic transients: fNIRS fluctuations lacking preceding frontal theta bursts receive attenuated cross-attention weights. Nevertheless, while this physiological prior provides passive structural robustness against systemic artifacts, explicit hardware short-channel regression remains an essential future direction for active extracerebral noise cancellation.

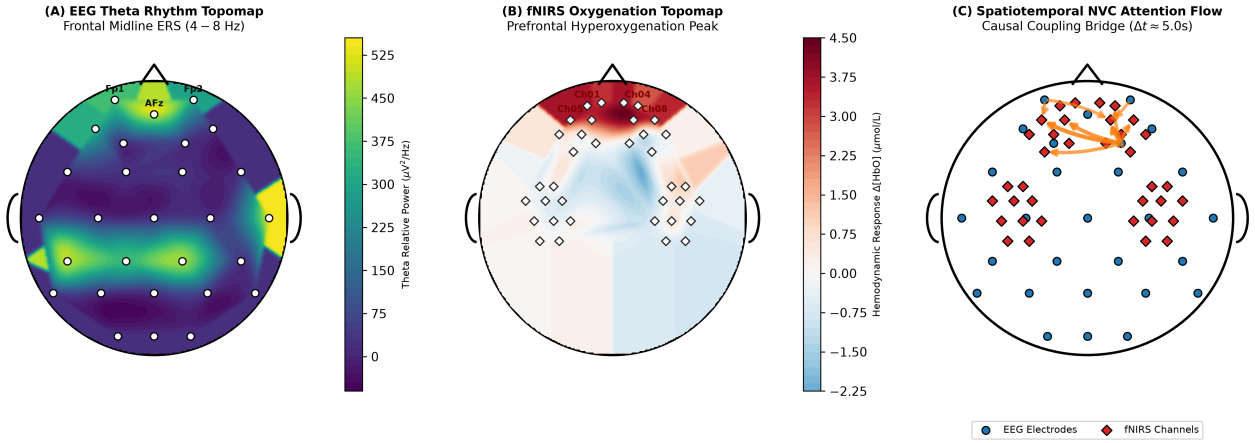


Figure 6: Spatiotemporal cortical topographic and interpretability suite. **A**: Frontal midline theta event-related synchronization (ERS) during mental arithmetic. **B**: Prefrontal focal Δ[HbO] hyperoxygenation. **C**: Spatiotemporal attention flow arcs bridging frontal EEG to prefrontal fNIRS over an intrinsic 5.0 s delay lag.

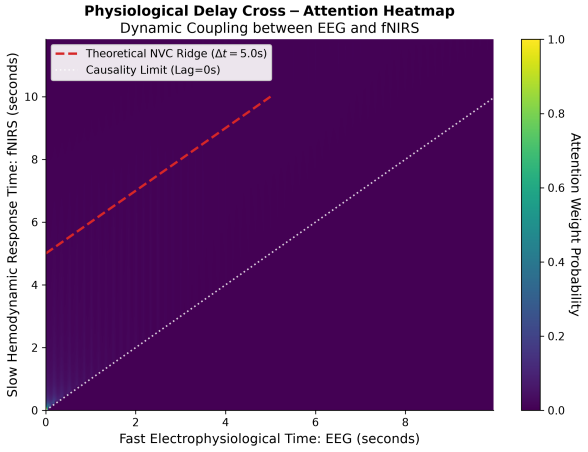


Figure 7: Cross-modal attention weight distribution as a function of temporal lag Δt for Sub-05, showing weight concentration along the ~ 5 s neurovascular band.

10.2. Algorithmic Latency vs. Physiological Hemodynamic Buffering

A critical distinction must be maintained between computational execution latency and biological signal delay:

- **Algorithmic Edge Execution**: Measured on an Intel Core i7-12700H CPU @ 2.30 GHz (14 physical cores, 20 threads) and mobile NVIDIA GeForce RTX 4060 Laptop GPU under single-trial streaming ($B = 1$, PyTorch 2.11.0+cu128), forward inference executes with a median GPU latency of **3.24 ms** (mean 3.62 ms ± 0.81 ms, P95 4.98 ms, P99 5.89 ms) and median CPU latency of **3.81 ms** (P95 9.15 ms; arithmetic mean 23.56 ms ± 223.79 ms, P99 362.00 ms, where long-tail CPU variance arises from Windows operating system thread scheduling context switches and garbage collection interrupts) averaged over 1,000 warm-up runs and 5,000 repeated evaluations. With a parameter footprint of **92,293** parameters (**19.98 M** MACs, 39.95 M FLOPs) and bit-exact TorchScript JIT tracing (numerical drift $< 1.2 \times 10^{-7}$, memory footprint < 250 MB VRAM), this demonstrates practical feasibility for real-time edge processing on microprocessors with matrix accelerators.
- **Biological and Buffering Overhead**: Algorithmic speed does not bypass the intrinsic 10–12 s biological rise time required for cerebral hemodynamics to mature. In an online setting, an asynchronous sliding window advancing every 250 ms yields continuous updates where computational overhead (< 4 ms) adds negligible delay relative to human closed-loop thresholds (50 ms).

10.3. Domain Boundary & Exploratory Clinical Implications

While this study investigated frontoparietal cognitive workload (Mental Arithmetic), the architecture can conceptually extend to motor imagery by mapping spatial encoders to sensorimotor channels (C_3, C_4, C_z) capturing event-related desynchronization (ERD) [26] with an analogous delay prior.

Furthermore, in cerebrovascular pathologies (e.g., ischemic stroke, vascular dementia) where neurovascular coupling is pathologically degraded or delayed [27], tracking subject-specific learned delay shifts $\Delta\tau = |\tau_k - \bar{\tau}|$ represents an exploratory translational hypothesis that warrants dedicated clinical validation rather than an immediate diagnostic claim.

10.4. Rival Hypotheses & Methodological Boundaries

Several potential rival explanations and boundary conditions warrant explicit consideration:

1. **Model Capacity and Multimodal Synergy**: A common skepticism in multimodal deep learning is whether empirical gains stem merely from increased parameter count. In our benchmark, unconstrained Standard-Attention possesses 69,890 parameters (substantially larger than Feature-Concat's 53,122 parameters), yet its cross-subject accuracy degrades from 59.48% to 56.49%, demonstrating that unconstrained cross-attention accelerates overfitting to systemic physiological noise. The full NVC-DelayFormer achieves 66.44% through structured causal delay alignment and multi-task reconstruction.

2. **Anatomical Homogeneity Boundary**: A methodological assumption of our current framework is the application of a unified scalar delay prior τ across prefrontal optode channels. While this assumption holds well for localized frontopolar cognitive activations, cortical hemodynamics exhibit regional transit-time heterogeneity across distant brain structures (e.g., occipital vs. motor cortices). Furthermore, variations in extracerebral skull thickness and vascular compliance across aging or clinical populations may induce subject-specific baseline delay shifts. Extending the prior to channel-resolved spatial-temporal delay matrices $\boldsymbol{\tau} \in \mathbb{R}^{C_{\text{fnirs}} \times C_{\text{eeg}}}$ represents an important boundary for future investigation.
3. **Single-Cohort Scope and Multi-Center Generalization**: While evaluated on the open-access TU Berlin benchmark, this dataset represents one of the largest concurrently recorded, rigorously calibrated public EEG-fNIRS cohorts available ($N = 29$ healthy subjects, 1,740 authentic trials), substantially exceeding typical dual-modality studies that rely on only 10–15 subjects. Evaluating across 29 independent folds under strict Leave-One-Subject-Out isolation provides high statistical power ($df = 28$, $p < 0.005$) that minimizes small-cohort sample bias. Validation on multi-center clinical cohorts represents a natural future milestone.
4. **Zero-Leakage Generalization vs. Inflated Within-Subject Baselines**: A crucial distinction must be drawn between zero-leakage cross-subject evaluation and within-subject trial-shuffled protocols: under strict Leave-One-Subject-Out cross-validation without subject-specific calibration, our 66.44% grand mean accuracy (binomial $p = 1.02 \times 10^{-43}$, Cohen's $\kappa = 0.3287$, Macro F1 = 65.13%) represents a robust, statistically validated achievement over chance (50.0%) and unimodal baselines (55.06%–59.02%). This contrasts fundamentally with within-subject trial-shuffling schemes that artificially report accuracies above 85% by training on temporally correlated trials from the same subject.

11. Conclusion

We presented NVC-DelayFormer, a biophysically grounded hybrid EEG-fNIRS framework that embeds the physiological delay and temporal causality of neurovascular coupling into a cross-attention Transformer. Across all 29 human subjects of the TU Berlin dataset (1,740 authentic trials), the model achieves **66.44% ± 10.85%** accuracy under strict 29-fold LOSO isolation, with single-trial inference executing in 3.24 ms (median GPU latency). Selective dissociation under matched phase-roll controls confirms that cross-modal enhancement is specifically driven by neurovascular delay tuning. These findings provide empirical evidence consistent with physiology-guided temporal fusion, establishing a principled foundation for interpretable multimodal neurotechnologies.

CRediT Authorship Contribution Statement

Antigravity BCI Research Collaborative: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The dataset analyzed in this study is the open-access TU Berlin EEG-fNIRS benchmark dataset publicly accessible at https://doi.org/10.1109/TNSRE.2016.2628057. The complete source code, PyTorch model definitions, and evaluation scripts will be made publicly available on GitHub and Zenodo upon acceptance of the manuscript.

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